Rotaract Club of Xiamen University Malaysia | Rotary Club of Kuala Lumpur Dynamics | Rotary Club of Tun Razak Exchange

SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

Instagram: @sdc_xmumhack ***Email:*** sdc.xmumhack@gmail.com

SDC HACKATHON 2026

Phase 1: Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club
Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

1. Team & Project Overview

Team Name	NovaMind
Selected Domain	Domain 1 – Aiversecity: Future Ready Workforce
Selected SDG Track (1–17)	SDG4, SDG8
Project / Solution Name	FuturePath AI
Name (Leader)	Christopher Gan Wei Hong
Email & Phone (Leader)	chgan775@gmail.com
University Affiliation(s)	Asia Pacific University (APU)
Number of Team Members	5

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

The rapid advancement of Artificial Intelligence is altering workforce demands at a faster pace than traditional academic curricula and static career guidance frameworks can adapt. As a result, undergraduate students, graduating seniors, fresh graduates, and early-career switchers face severe uncertainty regarding which traditional roles are becoming obsolete and which emerging AI-era competencies they need to develop. Without access to up-to-date market intelligence, individuals risk investing significant time and tuition into declining career paths. This issue is critical because it creates a widening structural skill gap across industries, severely impacting economic stability and the employability of tomorrow's talent.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

FuturePath AI is a web-based career guidance prototype designed to help university students and fresh graduates understand how their existing skills align with six technology careers.

Users select their skills and interests, then receive ranked career recommendations calculated using a Python-based weighted skill-alignment algorithm. The application explains each score by displaying matched skills, missing skills and the competencies users could develop next.

The platform also provides learning resources, personalised reskilling plans and downloadable learning plans. To provide job-market context, it displays a dated snapshot of remote job listings collected from the Himalayas API. Occupational skill benchmarks are supported by O*NET and additional career research.

The current prototype uses Python, Streamlit and web interface components, with deployment through Replit. Future development may introduce more comprehensive datasets, semantic CV analysis and advanced AI-powered career recommendations.

FuturePath AI aligns with SDG 4 (Quality Education) and SDG 8 (Decent Work and Economic Growth).

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

Our prototype is built using Python and Streamlit and is deployed on Replit. The platform features a Profile Intake module that captures a user's current technical and soft skills. Instead of complex LLM scoring, our matching engine utilizes a custom **weighted skill-alignment formula** to evaluate the user's profile against a static, dated snapshot of job listings (the Himalayas dataset). This allows the system to accurately calculate match percentage scores for existing roles and explicitly pinpoint current skill deficits.

Frontend UI & Hosting: Streamlit hosted on Replit

Backend / Logic: Python

Data Source (Current): Himalayas job-listing dataset (static/dated snapshot)

Matching Algorithm: Weighted skill-alignment formula

Architecture Flow: Frontend UI (Streamlit on Replit) -> Matching Engine (Python using weighted skill-alignment) -> Data Layer (Static Himalayas Dataset)

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

Our team applied design-thinking by prioritizing a functional, user-centric prototype that addresses the immediate need of undergraduate students and early-career switchers: identifying exact skill gaps. By deploying a lightweight Streamlit web application on Replit, we ensured rapid prototyping and immediate accessibility. We purposely constrained our initial MVP to use a static job-listing snapshot (Himalayas data) and a transparent weighted skill-alignment formula. This lean approach allows us to validate the core user experience—capturing profiles and identifying skill deficits—before engineering the more complex predictive models, live data pipelines, and adaptive improvements to our automated learning roadmaps planned for Phase 2.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	**Your Response**
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	• SDG 4 (Quality Education) - Target 4.4: Substantially increasing the number of youth and adults with relevant technical and vocational skills for employment and entrepreneurship. • SDG 8 (Decent Work & Economic Growth) - Target 8.5: Achieving full and productive employment and decent work for all women and men.

Metric Category	Your Response
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	• Currently, there is a widening structural skill gap across industries as individuals invest in declining career paths due to static academic curricula and lack of up-to-date market intelligence.
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	• Accuracy of skill-gap identification (measured by the weighted skill-alignment formula) and the number of users matched viable roles based on current snapshot data.
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	• Successfully transition the matching engine from static snapshot data to live predictive market data, and implement actionable, automated learning roadmaps for 100% of generated skill gaps.

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

- ☐ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- ☑ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- ☑ The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- ☑ All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☑ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☑ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
Streamlit	Code	Apache 2.0
O*NET datasets	Datasets	CC BY 4.0 (Creative Commons Attribution)
Himalayas datasets	Datasets	CC0 (Public Domain) / Custom Free Use

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☑ No

If yes, please describe the nature of the agreement and which member it concerns:

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.



[Signature]

24 October 2026

[Date]